

MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)

Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi
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“AN EXPLAINABLE RoBERTa-BASED MULTIMODAL DEEP LEARNING FRAMEWORK FOR ONLINE RECRUITMENT FRAUD DETECTION”

**A thesis report submitted in partial fulfillment of the requirements for
the award of degree of**

MASTER OF COMPUTER APPLICATIONS

Submitted by

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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that the thesis work entitled “AN EXPLAINABLE RoBERTa-BASED MULTIMODAL DEEP LEARNING FRAMEWORK FOR ONLINE RECRUITMENT FRAUD DETECTION” is a bonafide work carried out by P KARTHIK AITHAL (4MT24MC056), in partial fulfillment for the award of degree of Master of Computer Applications of Mangalore Institute of Technology & Engineering (Autonomous) during the year 2025-26. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The thesis has been approved as it satisfies the academic requirements in respect of Major Project Phase-2 work prescribed for the Master of Computer Applications degree.

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ABSTRACT

An Explainable and Robust DL framework (ERDL) is proposed in this thesis to detect and ease the ORF from electronic job market. This study aims to develop an accurate, robust and explainable classification system to differentiate real corporate job postings from well-established generalised fake job ads that closely resemble the actual corporate job postings. Since it uses deep semantic representations of both jobs and structured recruiter meta data to form a multi-modal feature set, the system will perform well. The extreme class imbalance and data insensitivity characteristic of Online Recruitment Fraud (ORF) datasets, which is another subgoal of this project, which is addressed using Synthetic Minority Oversampling Technique (SMOTE). The another subgoal of this project is solving the problems caused by the usual "black-box" issue of deep neural networks (DNNs) in cyber security through Explainable AI (XAI), which explains how a classification decision was made.

Directly aligned with these established goals, the observed results of this study demonstrate that the fine-tuned RoBERTa-based multi-modal classifier proposed in reached an exceptional overall detection accuracy of 98.4%, significantly surpassing commonly used machine learning classifiers (e.g., Random Forest and Support Vector Machines) as well as classic BERT architectures. These results are consistent with the justification that combining text embeddings with structural metadata drastically diminishes false-positive rates. Finally, SHAP (SHapley Additive exPlanations) joins hands well here and maps token-level contributions to the probability of being a fraud. The results show our classifier has learnt genuine semantic signatures of fraud requests (such as upfront fee asking, uplifted compensation and personal messenger app redirect (e.g. Telegram)) which are shown to the user in a human understandable visualisation.

Keywords: Online Recruitment Fraud (ORF), RoBERTa, Transformer, Deep Learning, Multi-Modal Detection, SMOTE, Explainable AI (XAI), Web Scraping, Natural Language Processing, Cybersecurity.

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